

JUNHUI PENG, PhD

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EDUCATION

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| Ph.D. | University of Science and Technology of China (USTC)
Computational Biology, School of Life Sciences
Advisor: Dr. Zhiyong Zhang & Dr. Yunyu Shi | Jul 2017 |
| B.S. | University of Science and Technology of China (USTC)
Biology, School of Life Sciences
Advisor: Dr. Zhiyong Zhang & Dr. Yunyu Shi | Jun 2012 |

RESEARCH EXPERIENCES

Postdoctoral Researcher, Rockefeller University Jul 2019 – present

Advisor: Dr. Li Zhao

- Developed a computational platform to study the origin and evolution of the interactions between sex peptide and sex peptide receptor. This demonstrates how genetic novelty gives rise to novel protein-protein interactions.
- Developed a computational pipeline to identify *Drosophila* specific *de novo* genes and study how genetic novelty gives rise to protein structure novelty.
- Demonstrated how intermolecular interactions drive protein sequence adaptive evolution in *Drosophila melanogaster* by comparative genomics analysis and population genomics analysis.
- Collaborated with Prof. Hironori Funabiki, Prof. Kivanç Birsoy, and other colleagues in comparative genomic analysis, population genomics analysis, and computational biology studies.

Postdoctoral Researcher, Hong Kong University of Science and Technology (HKUST) Oct 2017 – Jun 2019

Advisor: Dr. Xuhui Huang

- Applied molecular dynamics simulations to study the dynamics of nucleosome DNA unwrapping.
- Collaborated with Prof. Benzong Tang and other colleagues in computational biophysics studies.

Graduate studies, University of Science and Technology of China (USTC) Sep 2012 – Jun 2017

School of Life Sciences

Advisor: Dr. Zhiyong Zhang & Dr. Yunyu Shi

- Thesis: The development and application of computational platforms that integrates real world biophysical data to study the structure and dynamics of biomolecules.
- Collaborated with Prof. Ming Lei, Prof. Qingguo Gong, Prof. Congzhao Zhou, and other colleagues in computational biology.

Undergraduate studies, USTC Sep 2011 – Jun 2012

School of Life Sciences

Advisor: Zhiyong Zhang & Yunyu Shi

FUNDING & AWARDS

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|---|------|
| Shapiro-Silverberg Fund for the Advancement of Translational Research, The Rockefeller University
Center for Clinical and Translational Science (RUCCTS) | 2024 |
| SMBE Travel Award, Society for Molecular Biology and Evolution | 2023 |

C. H. Li Memorial Scholar Fund Award, Rockefeller University	2021
CAS President Award (Excellence Award), Chinese Academy of Science	2017

PRESENTATIONS

Oral, SMBE Satellite Meeting on De Novo Gene Birth, Texas A&M University, TX	Nov 2023
Oral, the 64 th Annual Drosophila Research Conference, Chicago, IL	Mar 2023
Oral, 2023 New York Area Population Genomics Meeting	Jan 2023
Oral, BSVMRKYZ Super-Group Meeting, Rockefeller University, NY	Nov 2022
Oral, Pels Family Center Chemical and Structural Biology Retreat, Edith Macy Center, NY	Oct 2022
Poster, virtual, the 63 rd Annual Drosophila Research Conference	Apr 2022
Oral, virtual, the 2 nd AsiaEvo Conference	Aug 2021
Oral, Chinese Society of Biochemistry and Molecular Biology Conference, Xiamen, China	Aug 2014

PUBLICATIONS (*more information at [my google scholar profile](#)*)

Postdoctoral research, Rockefeller University

1. Peng, J., Wang, B.-J., Svetec, N. & Zhao, L. Gene regulatory networks and essential transcription factors for de novo originated genes. **Nat Ecol Evol** 2025 (in press).
2. Wassing, I. E., Nishiyama, A., Shikimachi, R., Jia, Q., Kikuchi, A., Hiruta, M., Sugimura, K., Hong, X., Chiba, Y., **Peng, J.**, Jenness, C., Nakanishi, M., Zhao, L., Arita, K., Funabiki, H. CDCA7 is an evolutionarily conserved hemimethylated DNA sensor in eukaryotes. **Sci Adv**, 10(34), 2024
3. **Peng, J.**, Svetec, N., Molina N., Zhao L. *The origin and evolution of the interactions between sex peptide and sex peptide receptor*. **Mol Biol Evol** msae065, 2024
4. **Peng, J.**, Zhao, L. *The origin and structural evolution of de novo genes in Drosophila*. **Nat Commun** 15, 2024
5. Liu Y., Liu S., Tomar A., Yen F., Unlu G., Ropek N., Weber R., Wang Y., Khan A., Gad M., **Peng J.**, et al. *Autoregulatory control of mitochondrial glutathione homeostasis*. **Science** 382, 820-828, 2023
6. Chung, K., Xu, L., Chai, P., **Peng, J.**, Devarkar, S. C., Pyle, A. M. *Structures of a mobile intron retroelement poised to attack its structured DNA target*. **Science** 378, 627-634, 2022
7. **Peng, J.**, Svetec, N., Zhao, L. *Intermolecular Interactions Drive Protein Adaptive and Coadaptive Evolution at Both Species and Population Levels*. **Mol Biol Evol** msab350, 2022
8. Durkin, S.M., Chakraborty, M., Abrieux, A., Lewald, K.M., Gadau, A., Svetec, N., **Peng, J.**, Kopyto, M., Langer, C.B., Chiu, J.C., et al. *Behavioral and genomic sensory adaptations underlying the pest activity of Drosophila suzukii*. **Mol Biol Evol** msab048, 2021

Postdoctoral research, HKUST

9. Pan, C., Liu, C., **Peng, J.**, Ren, P., and Huang, X. *Three-site and five-site fixed-charge water models compatible with AMOEBA force field*. **J Comput Chem** 41, 2020
10. Zhang, J., Li, A., Zou, H., **Peng, J.**, Guo, J., Wu, W., Zhang, H., Zhang, J., Gu, X., Xu, W., et al. *A "simple" donor-acceptor AIEgen with multi-stimuli responsive behavior*. **Mater Horiz** 7, 2020
11. Zhang, J., Liu, Q., Wu, W., **Peng, J.**, Zhang, H., Song, F., He, B., Wang, X., Sung, H.H.Y., Chen, M., et al. *Real-Time Monitoring of Hierarchical Self-Assembly and Induction of Circularly Polarized Luminescence from Achiral Luminogens*. **ACS Nano** 13, 2019
12. **Peng, J.**, Wang, W., Yu, Y.Q., Gu, H.L., and Huang, X. *Clustering algorithms to analyze molecular dynamics simulation trajectories for complex chemical and biological systems*. **Chin J Chem Phys** 31, 2018

Graduate research, USTC

13. **Peng, J.#**, Yuan, C.#, Hua, X., Zhang, Z. *Molecular mechanism of histone variant H2A.B on stability and assembly of nucleosome and chromatin structures. Epigenetics Chromatin* 13, 2020 (**# co-first author**)
14. **Peng, J.#**, Yuan, C.#, Ma, R., Zhang, Z. *Backmapping from Multiresolution Coarse-Grained Models to Atomic Structures of Large Biomolecules by Restrained Molecular Dynamics Simulations Using Bayesian Inference. J Chem Theory Comput* 15, 2019 (**# co-first author**)
15. Xu, D., Ma, R., Zhang, J., Liu, Z., Wu, B., **Peng, J.**, Zhai, Y., Gong, Q., Shi, Y., Wu, J., et al. *Dynamic Nature of CTCF Tandem 11 Zinc Fingers in Multivalent Recognition of DNA As Revealed by NMR Spectroscopy. J Phys Chem Lett* 9, 2018
16. Cheng, P.#, **Peng, J.#**, and Zhang, Z. (2017). SAXS-Oriented Ensemble Refinement of Flexible Biomolecules. *Biophys J* 112, (**# co-first author**)
17. Chen, C., Gu, P., Wu, J., Chen, X., Niu, S., Sun, H., Wu, L., Li, N., **Peng, J.**, Shi, S., et al. *Structural insights into POT1-TPP1 interaction and POT1 C-terminal mutations in human cancer. Nat Commun* 8, 2017
18. Xu, L., Wang, L., **Peng, J.**, Li, F., Wu, L., Zhang, B., Lv, M., Zhang, J., Gong, Q., Zhang, R., et al. *Insights into the Structure of Dimeric RNA Helicase CsdA and Indispensable Role of Its C-Terminal Regions. Structure* 25, 2017
19. **Peng, J.**, Zhang, Z. *Unraveling low-resolution structural data of large biomolecules by constructing atomic models with experiment-targeted parallel cascade selection simulations. Sci Rep* 6, 2016
20. Shao, Z., Yan, W., **Peng, J.**, Zuo, X., Zou, Y., Li, F., Gong, D., Ma, R., Wu, J., Shi, Y., et al. *Crystal structure of tRNA m1G9 methyltransferase Trm10: Insight into the catalytic mechanism and recognition of tRNA substrate. Nucleic Acids Res* 42, 2014
21. **Peng, J.**, Zhang, Z. *Simulating large-scale conformational changes of proteins by accelerating collective motions obtained from principal component analysis. J Chem Theory Comput* 10, 2014
22. Wen, B.#, **Peng, J.#**, Zuo, X., Gong, Q., and Zhang, Z. *Characterization of protein flexibility using small-angle x-ray scattering and amplified collective motion simulations. Biophys J* 107, 2014 (**# co-first author**)

TEACHING & MENTORSHIP

Mentor for Sara Skarabot in single cell RNA-sequencing analysis <i>Current status: High school student intern at Stuyvesant High School</i>	2023
Mentor for lab members in comparative genomics and computational biophysics analysis <i>McDonald-Kreitman test, orthoMCL, dN/dS, AlphaFold, Protein structure analysis, etc.</i>	Since 2020
Mentor for Chuang Yuan in molecular dynamics (MD) simulations and data analysis (master's thesis) <i>Current status: Research assistant at Shandong University</i>	2016
Mentor for Zhiyuan Ding in MD simulations of nucleosome assembly (undergraduate thesis) <i>Current status: PhD student at Fudan University</i>	2016
Mentor for lab members in biophysical data analysis <i>Small-angle X-ray Scattering (SAXS), NMR spectroscopy data analysis, Structural modeling integrating biophysics data</i>	2014-2017
Teaching assistant, Molecular Modeling of Biomacromolecules at graduate level	2014

SERVICE & OUTREACH

Faculty Search Ambassador Navigate candidates through campus and introduce them to faculty members and resource centers, Rockefeller University Faculty Search Seminar 2023	2023
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Science Networking Host	2022
Host of virtual networking at 2022 Annual Drosophila Meeting, session of Evolution, Immunity, and the Microbiome	
Ad Hoc Reviewer	Since 2018
Molecular Biology and Evolution, PLOS Genetics, Genome Biology and Evolution, Journal of Molecular Evolution, Journal of Evolutionary Biology, Biology of Reproduction, PLOS One, Archives of Biochemistry and Biophysics, International Journal of Molecular Science, BioSystems, Computational Biology and Chemistry, Journal of Molecular Structure	
Organizer of the first Biological System Modeling Symposium	2015
Organizer of the first campus-wide symposium that connects scientists who study biological systems by computational modeling and simulations in USTC	
Science Outreach Volunteer	2014, 2015, 2016
Introduce science and provide hands-on scientific experiments to elementary, middle, and high school students from all over the country in the Science and Technology Activity Week, School of Life Sciences, USTC	

PROFESSIONAL MEMBERSHIP

Society of Molecular Biology and Evolution	Since 2020
Genetic Society of America	Since 2020